STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Integration — Subject Rules, Secure Client–Server Communication, Session Tracking & Concurrency]	TEAM ID : **2** FACULTY MENTOR: Mr. Rudragouda Patil	WEEK ENDING: 31/08/2026 STUDENT LEADER: Kartik Manoj.

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Subject Creation → /api/subjects:** Traced the Bearer-token flow: login issues a token (SessionManager), sessionSecurity.js attaches the Authorization header, and AppServer checks token and role on only two endpoints (/api/admin/users, /api/quiz-results/grade); security headers are set on all responses. • **Self-Moderation Rule:** Reviewed moderator.html: review actions are blocked when the question bank's author email or name matches the logged-in faculty member. • **Authenticated Communication:** Traced the Bearer-token flow (SessionManager → sessionSecurity.js → AppServer role check). Reviewed apiClient.js (retry, /api/health heartbeat, HTTP 429 handling); no page loads it yet. • **Login Limit & Session Time:** Traced LoginRateLimiter (5 failed attempts, 15-minute lockout, HTTP 429) and the logout → sessionActiveTime → audit log → admin "Session Active Time" column. • **Concurrency Scripts:** Reviewed simulate_concurrent_authors.py (5 authors, one subject, under SubjectConcurrencyManager lock) and simulate_100_quiz_students.py (POST /api/quiz-results).	• **No Transport Encryption:** AppServer is plain HTTP (HttpServer), although an HSTS header is sent; CORS allows all origins; demo_admin_token / demo_author_token are accepted as valid tokens; all other endpoints need no token. • **Rules Enforced Only in Browser:** Self-moderation, subject-code format and auto-versioning run in the browser only; /api/question-banks has no auth or ownership check, and /api/subjects accepts any code (defaults to SUB01). • **Session Time & Client Module:** Session time is computed in the browser at logout only; apiClient.js is unused, so retry and heartbeat are inactive. • **50-Client Test Not Evidenced:** Script only POSTs final results (140 threads); no ongoing-test handling. Summary claims 100/100 but no output is saved. • **Tests Not Run:** Lockout is per login ID, not per device; lockout and concurrent-author tests have no recorded run.	• Need a decision from the mentor and Backend team: is HTTPS required for the final demo, or is HTTP on localhost acceptable? If required, Backend must supply the keystore and HTTPS server setup before end-to-end testing.

INTEGRATION AND COLLABORATION	COMMON KPI EVIDENCES
Interfaces Used: POST /api/subjects (assignedAuthors, assignedModerator, version), POST /api/auth/login (429 on lockout), POST /api/auth/logout (sessionActiveTime), GET /api/audit-logs, POST /api/question-banks, POST /api/quiz-results; Authorization: Bearer header. **Components Received (not developed by Integration):** • **Backend team —** AppServer, SessionManager, LoginRateLimiter, SubjectConcurrencyManager. • **Frontend team —** Admin subject form, moderator.html, login.html, sessionSecurity.js, apiClient.js. **Open Dependencies:** HTTPS decision; server-side enforcement of subject and moderation rules; 50-client test. **Pipeline Learnings:** Login → token → Bearer calls → logout → audit log → admin dashboard; several business rules currently live only in the browser.	• **Foundational Behavior:** Lockout after 5 failed attempts for 15 minutes; server session expires after 2 h idle; browser warns after 15 min inactivity. No latency/CPU figures recorded. • **Continuous Integration:** Test scripts exist in integration/ (test_api_endpoints.py, simulate_*.py, test_security_audit_module7.js); none has a recorded run yet. • **Instrumentation:** Audit-log entries carry session active time and subject version changes.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Shruti Chavan:** Run and record the 5-author concurrent test and a lockout check; raise server-side rule enforcement. • **Kartik Manoj:** Agree HTTPS approach with Backend and decide whether pages adopt apiClient.js. • **Anagha Thavkar:** Run a 50-client simulated test and record success count and latency. • **Pratiksha Chavan:** Document auth, lockout, session and subject endpoints plus test results in the integration README.	